

Research Progress — Report 3

Multilingual LLM Simulation of Human Decision-Making Diversity: A Japanese–Vietnamese Cross-Cultural Study

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Overview

1. **Progress** — Past 2 weeks
2. **Paper reading findings** — What we learned & relevance to our research
3. **AI interview design** — Current plan & questions for Prof. Tho
4. **Plan** — Next 2 weeks

1. Literature Review

- Conducted a reading session of **2 foundational papers** on the Foreign Language Effect (FLE)
 - Keysar, Hayakawa & An (2012) — *Psychological Science*
 - Costa, Foucart, Hayakawa et al. (2014) — *PLOS ONE*
- Both papers serve as the **theoretical foundation** for our research

2. AI Interview Design

- Started brainstorming the **themes and decision tasks** for the AI interview
- Considering culturally appropriate and sensitive topic selection

Paper Reading Findings

Summary of the 2 papers

Paper	Research Question	Key Finding
Foreign Language Effect Keysar et al. (2012), <i>Psych. Science</i>	Does using a foreign language reduce decision biases?	FLE confirmed: framing effect and loss aversion significantly reduced in a foreign language; more analytic, less emotion-driven decisions
Your Morals Depend on Language Costa et al. (2014), <i>PLOS ONE</i>	Does language (L1 vs. L2) affect moral judgment?	More utilitarian choices in a foreign language in the footbridge dilemma; attributed to reduced emotional response

Relevance to our research: These two papers establish FLE in *human* participants. Our research asks the next question — does the same effect appear when LLMs *simulate* human decision-making? The decision and moral dilemma tasks used here are also useful references for our AI interview design.

AI Interview Design & Questions for Prof. Tho

Current direction

- **Avoiding** politically and historically sensitive topics (out of cultural sensitivity)
- Current candidate theme: Multicultural co-learning / Intercultural collaborative learning
 - Relatable for both Japanese and Vietnamese university students
 - Neutral — avoids historical or political sensitivities

Questions for Prof. Tho

1. Is **multicultural co-learning** a suitable and natural theme for Vietnamese students?
2. What **decision scenarios** would you recommend around this theme?

Plan — Next 2 Weeks (Hieu)

Literature reading & AI interview theme proposal

- Read the following papers and propose AI interview themes
 - FLE foundations: Circi et al. (2021) — *Meta-analysis of FLE in decision-making*, PB&R
 - Cultural Frame Switching: Ramírez-Esparza et al. (2006) — *Do bilinguals have two personalities?*, JRP
 - Adaptive AI interviewing: Wuttke et al. (2024) — *AI Conversational Interviewing*, arXiv:2410.01824
- Additional papers to be read to deepen understanding

Plan — Next 2 Weeks (Eita)

Literature reading & interview preparation

- Read the following papers
 - Vocal / confidence quantification: Eyben et al. (2015) — *eGeMAPS*, IEEE T. Affective Computing
 - Group optimization (applied background): Min. Entropy Collaborative Groupings (2023) — PLOS One
- Collect insights on **culturally appropriate interview topics**
→ Compile and submit a list of **candidate interview themes**
- Verify that **Gemini Live API** supports natural conversation in Japanese, English, and Vietnamese

Thank You

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